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MIL-M-69A
19 March 1971
SUPERSEDING
JAN-M-69
31 July 1944
(See 6.3)

MILITARY SPECIFICATION

MATS, CARGO

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers square type cargo mats.

1.2 Classification. Mats shall be of the following sizes, as specified (see 6.2).

Size 1 - 6 by 6 feet.
Size 2 - 4 by 4 feet.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of the specification to the extent specified herein.

SPECIFICATION

FEDERAL
T-R-650 - Rope, Yarn and Twine, Blast Fiber.

STANDARDS

MILITARY
MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-414 - Sampling Procedures and Tables for Inspection by Variables for Percent Defective.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

UNIFORM CLASSIFICATION COMMITTEE
Uniform Freight Classification Rules.

(Application for copies should be addressed to the Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Cargo mats shall be made from strands of 3-inch coir rope, or from 3-inch manila rope uniformly and firmly laid (see 6.2). The coir material shall be from Superior Anjengo, Superior Aratory, or Real Alapat Coir yarns.

3.2 Tolerance. A tolerance in the rope circumference of plus or minus 1/4 inch will be permitted.

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3.3 Construction. The strands of 3-inch coir rope or 3-inch manila rope, which has been unlaid, shall be hung over a jackstay of 21-thread ratline, conforming to type II, class 1 of T-R-650, and laid into a paunch mat. The bottom of the mat shall be finished over a second jackstay of 21-thread ratline, conforming to type II, class 1 of T-R-650, laid between the ends of the strands. One strand of each pair shall then be carried around the jackstay and laid up in place of the other strand of the pair, which is unlaid for about three turns. The ends shall then be tied in a half knot and tucked in so as to conform to the layers of the mat. The method of laying up the cargo mat and the general construction shall be as shown on figure 1.

3.4 Loops. Cargo mats shall have a 6-inch eye splice worked in both ends of each jackstay to facilitate handling the mat.

3.5 Thrums. Cargo mats shall be thrummed on the surface with strands of 3-inch coir rope or 3-inch manila rope. The thrums shall be inserted through the approximate center of two ropes at the point where they intersect (see figure 2), and shall project approximately 4 inches from the surface of the mat. The thrums shall be located at alternate intersections in each row. The position of thrums, and rows so thrummed, shall be staggered to produce even coverage of the cargo mat.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Sampling.

4.2.1 Lot. All cargo mats of the same size and produced in the same shop, using coir or manila from the same source, shall be considered a lot for purposes of sampling and examination.

4.2.2 Sampling. Sample mats shall be selected at random from each lot in accordance with MIL-STD-105 at inspection level II, the A.Q.L. shall be 2.5 percent.

4.3 Classification of defects. Defects shall be classified in accordance with table I.

Table I - Classification of defects.

Critical defects: None defined.	
Major defects:	
101	Mat is not the size specified.
102	Construction is not as specified.
103	Strands are not from 3-inch coir rope or 3-inch manila rope.
104	Jackstays are not 21-thread ratline conforming to T-R-650.
105	Ends of strands are not carried around the jackstay; are not tied in a half knot.
106	Eye splice is not worked in both ends of each jackstay; size of eye splice is not 6 inches.
107	Surface of the mat is not thrummed with 3-inch coir rope or 3-inch manila rope; thrums do not project 4 inches (approximately) from the mat surface.
108	Rope circumference does not conform to the specified limits.
Minor defects: None defined.	

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4.4 Examination.

4.4.1 Weight. Each of the sample mats selected in accordance with 4.2.2 shall be weighed separately, and those individual weights shall be used as shown in MIL-STD-414, section B, first to compute the standard deviation, and then to establish limits at 2 times the standard deviation above and below the mean weight. Mats which weigh outside those limits shall be rejected because they are not uniformly and firmly laid (see 3.1), or are not made of coir or manila from the same source lot (4.2.1).

5. PREPARATION FOR DELIVERY

5.1 Packing and marking for shipment.

5.1.1 Packing. Unless otherwise specified in the contract or order the subject commodity shall be shipped as bundles or bales securely formed by suitable metal strapping (flat or round) or rope. Bundles or bales shall conform to the requirement of the Uniform Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.1.2 Marking. Shipment marking information shall be provided on interior packages and exterior shipping containers in accordance with the supplier's commercial practice. The information shall include nomenclature, Federal stock number or manufactuere's part number, contract or order number, contractor's name and destination.

6. NOTES

6.1 Intended use. Cargo mats are intended for use as cargo cushions.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Size required (see 1.2).
(c) Type of rope required (see 3.1).
(d) Packing and marking (see 5.1).
(e) Quantity of items required.

6.3 CHANGES FROM PREVIOUS ISSUE. THE OUTSIDE MARGINS OF THIS DOCUMENT HAVE BEEN MARKED "#" TO INDICATE WHERE CHANGES (DELETIONS, ADDITIONS, ETC.) FROM THE PREVIOUS ISSUE HAVE BEEN MADE. THIS HAS BEEN DONE AS A CONVENIENCE ONLY AND THE GOVERNMENT ASSUMES NO LIABILITY WHATSOEVER FOR ANY INACCURACIES IN THESE NOTATIONS. BIDDERS AND CONTRACTORS ARE CAUTIONED TO EVALUATE THE REQUIREMENTS OF THIS DOCUMENT BASED ON THE ENTIRE CONTENT AS WRITTEN IRRESPECTIVE OF THE MARGINAL NOTATIONS AND RELATIONSHIP TO THE LAST PREVIOUS ISSUE.

Custodians:
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Navy - SH
Air Force - 82
Review activities:
Navy - SH
Air Force - 82
User interest:
Army - ME

Preparing activity:
Navy - SH
(Project 2040-0100)

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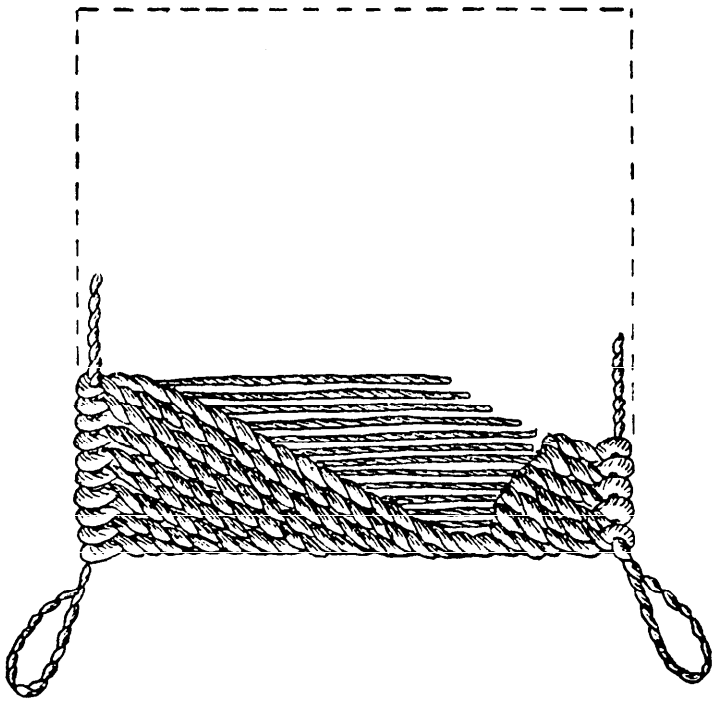


Figure 1 - Cargo mat.

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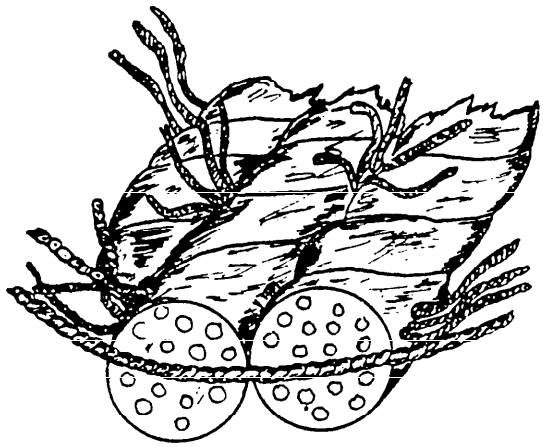


Figure 2 - Thrumming.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity (as indicated on reverse hereof).

SPECIFICATION

ORGANIZATION *(Of submitter)* | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT
$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD, FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4863, WHICH IS OBSOLETE

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